

Database Life Cycle

Topic 3

LOs

- By the end of this topic, you should be able to:
 - List the three designs involved in the database design phase;
 - State three important sections in the conceptual design phase; and
 - Explain the purpose of the six phases involved in the database development life cycle.

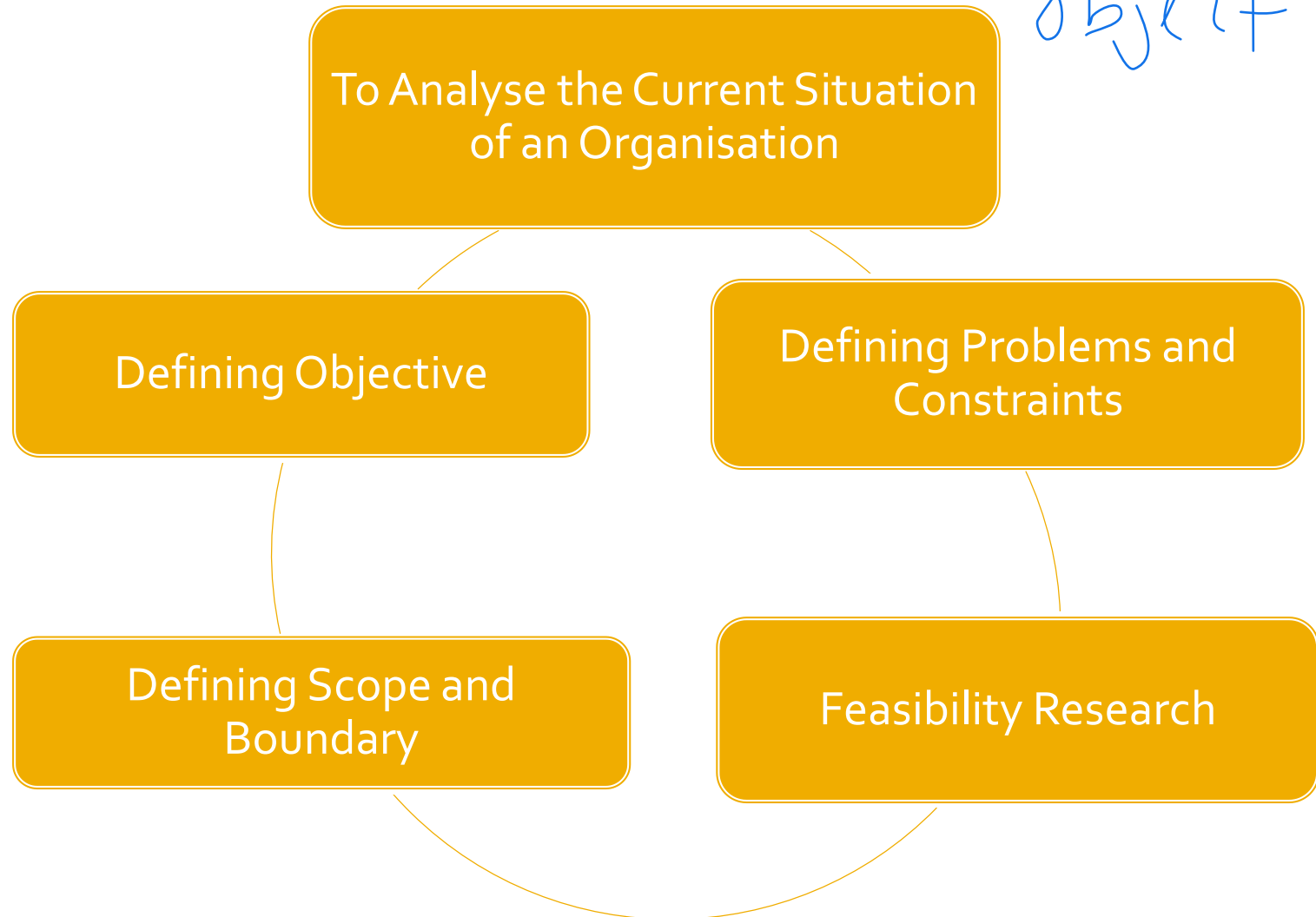
DDLDC

- A database system is a component of information system and is governed by one life cycle called Database Development Life Cycle (DDLDC).
- DDLDC is based on data orientation approach.
- The characteristics of data become the basis for designing the database scheme.
- All system functions are developed based on the scheme or database structure.
- The functions of a system changes according to the requirements of users but the database scheme usually remains stable without major changes.

DDLC

- The life cycle of a database development consists of six phases as listed below.
- These phases are iterative processes where each phase may be repeated a number of times in order to obtain a comprehensive feedback, especially from users.
 - (i) Database Planning
 - (ii) Database Design
 - (iii) Implementation and Downloading
 - (iv) Testing and Evaluation
 - (v) Operation
 - (vi) Maintenance and Evolution

1- DATABASE PLANNING



2-DATABASE DESIGN

- The design phase is the most important stage in a database development life cycle.
- It influences the whole performance of the database being developed.
- The purposes of a database design phase are as follows:
 - Represent data and the relationship between data.
 - Provide a data model to support all required transactions.
 - To determine that the system design has fulfilled the requirements of users and the system.

Conceptual Design

- **Conceptual Design** *fasāy ke3*
- Data modelling is used in conceptual design
- To design a database structure which can present real world objects in a manner which is realistic.
- The conceptual design is divided into three sections i.e. data modelling, normalisation and (data model validation) *y kita analyze hai*

chptr 6

Data Modelling

(a) Data Modelling

- The following questions must be answered before data modelling can be performed. Among the questions are:
 - (i) Information Requirement
 - What is the form or shape of the information output required from the system?
 - (ii) Information Source
 - Where can specific information be found and how do we obtain them?
 - (iii) Information Content
 - What are the characteristics of the data required to produce information such as attribute, size, number, data frequency and relationship between data? *apa hubungan dia*
 - (iv) Information User
 - Who will make use of the data? What are the views of users required?
siapa y akan tgk data-doctor *apa y ditgk*

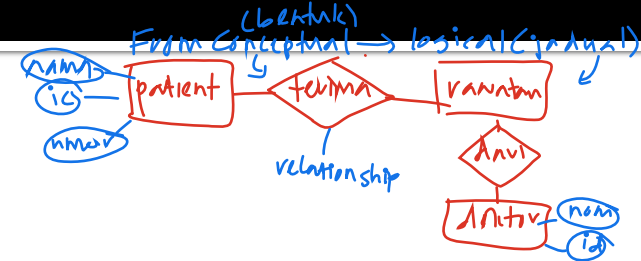
Normalisation

- (b) Normalisation → xnk data berulang → data y sama
 - A model being constructed must be flexible and accommodative i.e. able to suit changes according to operational changes including current and future organisational expansion.
 - The normalisation process determines the dependency of attributes to an entity and fixes relationships between entities through common attributes.
 - The normalisation process is implemented to overcome problems of data duplication, additional anomalies, updating and deletion which will be discussed in Topics 7 and 8.

Data Model Validation

- Data Model Validation *Satisfied semua keperluan data + tidak*
 - Models produced at the data modelling stage and at the normalisation stage may require changes or adjustments more than once.
 - The E-R diagram must be tested on the system process and functions proposed to confirm that the conceptual design can support the functions mentioned.
 - The conceptual model is free from DBMS model and the physical storage structure.

Logical Design



■ Logical Design

- The mapping of a conceptual design into a logical design means translating each entity, attribute and relationship into a data representation form compatible with a DBMS model.
- If the chosen DBMS data model is based on a relationship model, all objects in the conceptual data model will be changed into table forms consisting of rows and columns.

Physical Design

■ Physical Design

- The physical design involves physical aspects of data saved in storage and the performance of the system. The physical model is dependent on the DBMS software. Storage features depend on hardware whereas access features depend on the database system and DBMS.
- Now physical design is taken over by most DBMS to protect it from users (programmers).

3-IMPLEMENTATION AND DOWNLOADING

→ nak kena setup DBMS.

- Activities of the implementation phase and downloading are as follows;
 - (a) Setting Up a DBMS
 - A newly bought DBMS must be set up and users need to adapt themselves to the DBMS before it can be tested. A Database Administrator (DBA) will create database storage groups.
 - (b) Creating a Database
 - The database scheme, data dictionary and user's views are created through the use of DDL statement. This process is the same as giving permission to users to access specific table space.
 - (c) Data Downloading/Changing
 - Once a database is constructed, data will be downloaded into the database. This step is necessary only if the new database is constructed to replace the old database. Current data must be changed to the new data format.

4-TESTING AND EVALUATION

- (a) Database Testing and Application Procedures
 - The process of database testing is composed of integration test, unit test and system test.
- (b) Database Evaluation
 - The evaluation to be conducted will be on performance and system security. A database has to be protected from access by unauthorised users. DBMS kita manage
 - Steps which can be taken include tightening the physical security control, password, access rights, data confidentiality and simultaneous accessibility control.

5-OPERATION

- The operation phase and support system begins when the system has operated in an actual environment and ends when it is replaced by a new system.
- Once the system is established, the analyst must prepare the user's manual, a formal training session, technical support and create the centralisation of information.

Kita jg buat system ni kita kena ajar org tu how to use.

5-OPERATION

- A new system can operate by using several approaches as follows:
- Direct Transition Plan *System lama → baru.*
 - This operation involves sudden changes from the current system to the new system.
- Parallel Transition Plan
 - Both system operations are performed simultaneously until a certain period of time.
 - Once the management is satisfied with the new system, the existing system will be shut down.

5-OPERATION

■ Pioneer Transition Plan

- The new complete system is used only in a particular section or department of the company.
- At the same time, the existing system is still operational.
- If the running of the new system is smooth, the whole new system will be implemented in all sections.

■ Staggered Transition Plan

↳ dia berpevingkat.

- The new system is implemented according to phases or stages.
- If there are no problems in the system, the whole new system will be applied.

6-MAINTENANCE AND EVOLUTION

- Maintenance is the process of monitoring and maintaining the performance of the system and database.
- In performance monitoring, if the system is at an unsatisfactory level, reorganisation and readjustment of the database need to be carried out.

6-MAINTENANCE AND EVOLUTION

- Maintenance must be conducted from time to time for the purposes of:
 - (i) Correcting errors in the system, system competency and other factors that were not taken into consideration during the analysis stage but realised only later.
 - (ii) Several adjustments to the database shall be implemented when there are changes to the rules and organisational policy.
 - (iii) At certain stages, the system needs to be changed or upgraded to the latest version to support the needs and requirements of users as well as the organisation.

6-MAINTENANCE AND EVOLUTION

- However, these changes must not affect other users who use the same database.
- All problems must be attended to and solved by the DBA. A competent DBA shall have a maintenance chart ready and will not have to wait for a problem to surface before attending to it.

SUMMARY

- In conclusion, we have understood that there are six main phases in the database development life cycle.
- Each phase has its own function and purpose and the result of each phase is interrelated and valuable to the next phase.
- Generally, this life cycle is the process of developing a database beginning with planning, designing, implementation until the stage of maintenance.
① planning object.
② Design → conceptual
③ logical
- At the database planning stage, all information requirements such as finance and tasks to be carried out must be listed clearly and accurately.

SUMMARY

- Sound and detailed planning is needed to ensure smoothness when the process is launched and will result in a database that is efficient.
- In the database design phase, there are several other design levels i.e. conceptual design, logical and physical design.
- After this, the implementation and data downloading stages are carried out. Among the activities involved at this stage is the setting up of DBMS, creating a database and downloading or changing data.

SUMMARY

- Then the database system is tested and evaluated.
 - After going through the testing phase, the database enters the operational phase.
 - In this phase, the database system will be implemented in an actual environment.
- The last phase is the maintenance phase.
 - At this stage, the DBA must always carry out maintenance work to ensure that the database functions efficiently and without problems.